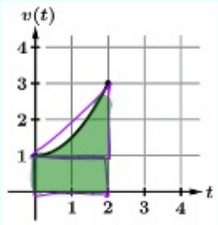
LIM-5.A.1 Definite integrals can be approximated for functions that are represented graphically, numerically, analytically, and verbally.
LIM-5.A.2 Definite integrals can be approximated using a left Riemann sum, a right Riemann sum, a midpoint Riemann sum, or a trapezoidal sum; approximations can be computed using either uniform or non-uniform partitions.
LIM-5.A.3 Definite integrals can be approximated using numerical methods, with or without technology.
LIM-5.A.4 Depending on the behavior of a function, it may be possible to determine whether an approximation for a definite integral is an underestimate or overestimate for the value of the definite integral.

6-2 Approximating Areas with Riemann Sums

Get Ready...

A particle Q is moving along a straight path, but its velocity is increasing as it moves. A portion of the graph of $v_Q(t)$ is given at the right where $v_Q(t) = 0.5t^2 + 1$. Approximate the total distance the particle traveled from $t = 0$ to $t = 2$ seconds.



Graph of $v_Q(t)$

$2 + 2 = 4$

Is your answer very accurate?

Although there are several methods to approximate area under a curve, we will use rectangles (Riemann sums) or trapezoids in AP Calculus.

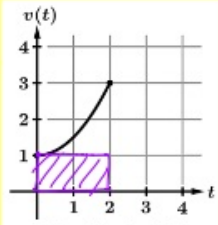
Approximating Area with Riemann Sums (rectangles)
When doing a **Riemann Sum** approximation (using **rectangles**) to approximate the area under a curve, we must decide these two things before starting our approximation: 1.) How many rectangles (or **subintervals**) will be used? 2.) Which method will we use for the height of each rectangle: **left endpoint**, **right endpoint**, or **midpoint**?

Example 1: Use a **left** Riemann sum with 1 subinterval to approximate the area between the curve $v_Q(t)$ and the t axis from $t = 0$ to $t = 2$.

base × height

left

left hand endpoint of the subinterval to find the height



Graph of $v_Q(t)$

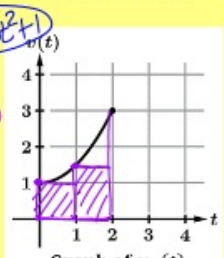
Area

$L_1 \approx 2 \cdot V_Q(0)$

$= 2(1) = 2$

Example 2: Use a **left** Riemann sum with 2 equal subintervals to approximate the area between the curve $v_Q(t)$ and the t axis from $t = 0$ to $t = 2$.

$[0,2]$ interval $V_Q(t)=0.5t^2+1$



Graph of $v_Q(t)$

Subinterval has a width of

$\Delta x = \frac{b-a}{n}$ n = # rectangles (subintervals)

$[a,b]$ interval

$\Delta x = \frac{2-0}{2} = 1 \leftarrow$ base

$V_Q(t)=0.5t^2+1$

$V_Q(0)=1$

$V_Q(1)=0.5+1=1.5$

Area

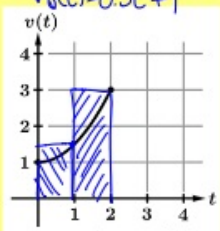
$L_2 \approx 1 \cdot V_Q(0) + 1 V_Q(1)$

$= 1(1)+1(1.5)$

$= 2.5$

Example 3: Use a **right** Riemann sum with 2 equal subintervals to approximate the area between the curve $v_Q(t)$ and the t axis from $t = 0$ to $t = 2$.

$V_Q(t) = 0.5t^2 + 1$



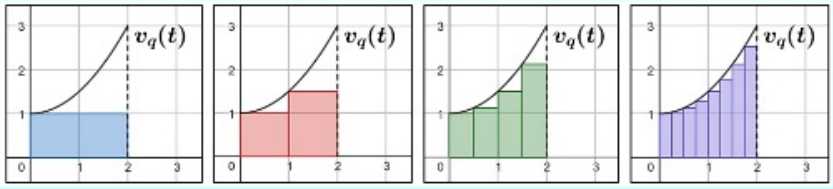
Graph of $v_Q(t)$

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{2} = 1$$

Area

$$R_2 \approx 1(V_Q(1)) + 1(V_Q(2))$$
$$= 1(1.5) + 1(3)$$
$$= 4.5$$

The more rectangles we use, the better our approximation (generally) becomes.



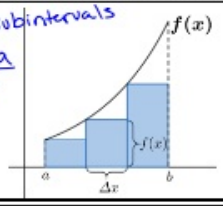
To find the exact area, we would need ____infinitely_____ many rectangles!

Connecting Riemann Sums to Calculus

Riemann Sum Approximation

EQUAL subintervals

$\Delta x = \frac{b-a}{n}$

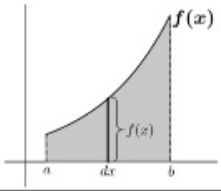


Number of Rectangles: n

Area of Each Rectangle: $f(x)\Delta x$

Total Area: $\sum_{k=1}^{n} f(x)\Delta x$

Exact Area



Number of Rectangles: ∞

Area of Each Rectangle: $f(x)dx$

Total Area: $\sum_{k=1}^{\infty} f(x)dx$

Notation: We will write $\int_a^b f(x)dx$ to represent the true area found by using infinitely many rectangles.

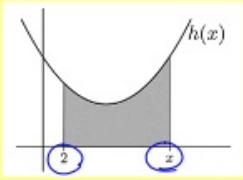
$\int_a^b f(x)dx$ is an **ACCUMULATOR** ... we are adding up (signed) areas of rectangles!

Example 4: Write an expression involving a definite integral that represents the shaded area in the figure at the right.

$$\int_2^x h(t)\,dt$$

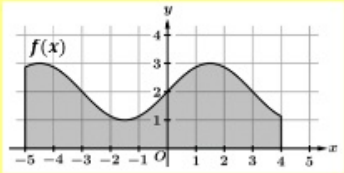
$h(x)$

Can't use x in the integral



Example 5: The graph of $f(x)$ is shown at the right. Write a definite integral that represents the shaded area.

$$\int_{-5}^{4} f(x)\,dx$$

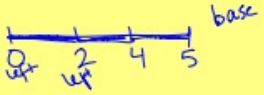


Approximating and Interpreting Areas

A large department store opens at 9AM and closes at 9PM. The rate that people enter the store, in people per hour, is modeled by the function $P'(t)$ where t is the number of hours since 9AM. The table above contains values of $P'(t)$ at selected times t. Use the data in the table above to answer the following.

t (hours)	0	2	4	5	6	9	12
$P'(t)$ (ppl/hour)	11	30	54	26	21	44	0

Example 6: Use a left Riemann sum with three subintervals to approximate the value of $\int_0^5 P'(t)dt$. Include units of measure.

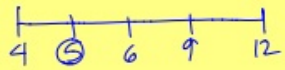


$$\int_0^5 P'(t)dt \approx 2 \cdot P'(0) + 2P'(2) + 1P'(4)$$
$$= 2(11) + 2(30) + 1(54)$$
$$= 22 + 60 + 54$$
$$= 136 \text{ people}$$

A large department store opens at 9AM and closes at 9PM. The rate that people enter the store, in people per hour, is modeled by the function $P'(t)$ where t is the number of hours since 9AM. The table above contains values of $P'(t)$ at selected times t. Use the data in the table above to answer the following.

t (hours)	0	2	4	5	6	9	12
$P'(t)$ (ppl/hour)	11	30	54	26	21	44	0

Example 7: Use a right Riemann sum with the four subintervals $[4, 5]$, $[5, 6]$, $[6, 9]$, and $[9, 12]$ to approximate the value of $\int_4^{12} P'(t)dt$. Include units of measure.

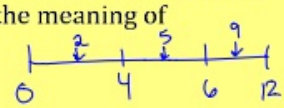


$$\int_4^{12} P'(t)dt \approx 1 \cdot P'(5) + 1 \cdot P'(6) + 3\,P'(9) + 3P'(12)$$
$$= 1(26) + 1(21) + 3(44) + 3(0)$$
$$= 26 + 21 + 132 + 0$$
$$= 179 \text{ people}$$

A large department store opens at 9AM and closes at 9PM. The rate that people enter the store, in people per hour, is modeled by the function $P'(t)$ where t is the number of hours since 9AM. The table above contains values of $P'(t)$ at selected times t. Use the data in the table above to answer the following.

t (hours)	0	2	4	5	6	9	12
$P'(t)$ (ppl/hour)	11	30	54	26	21	44	0

Example 8: Approximate the value of $\int_0^{12} P'(t)dt$ using a midpoint Riemann sum with three subintervals as indicated in the table above. Using correct units, interpret the meaning of $\int_0^{12} P'(t)dt$ in context of the problem.



$$\int_0^{12} P'(t)dt \approx 4 \cdot P'(2) + 2P'(5) + 6P'(9)$$
$$= 4(30) + 2(26) + 6(44)$$
$$= 120 + 52 + 264 = 436 \text{ people}$$

$\int_0^{12} P'(t)dt \approx 436$ people

436 people entered the store between 9Am + 9pm.

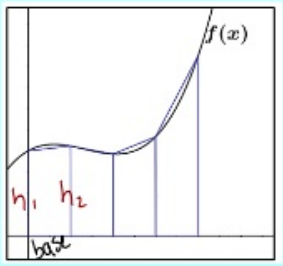
Trapezoidal Approximations

We have been using rectangles to approximate the area under a curve. However, this is not the only method that we can use to approximate the area or accumulation. Finding the area of rectangles is easy to calculate, which is why it is a good method to use.

Another shape that is relatively easy to use is the TRAPEZOID. Trapezoids are easy to use to find area, and they often result in a better approximation.

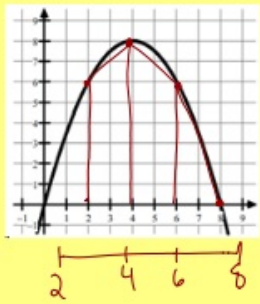
Area of a Trapezoid:

$$\frac{1}{2}(h_1+h_2)b$$



Example 9: The graph of $f(x) = 4x - \frac{1}{2}x^2$ is at the right. Use a trapezoidal approximation with 3 equal subintervals to approximate the area under the curve on the interval $[2, 8]$.

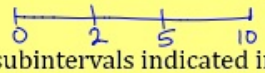
$$\Delta x = \frac{b-a}{n} = \frac{8-2}{3} = \frac{6}{3} = 2$$



$$\int_2^8 f(x)\,dx \approx \frac{1}{2}\left[f(2)+f(4)\right]\cdot 2$$
$$+ \frac{1}{2}\left[f(4)+f(6)\right]2$$
$$+ \frac{1}{2}\left[f(6)+f(8)\right]2$$
$$= 6+8+8+6+6+0$$
$$= 34$$

t (hours)	0 (9am)	2	5	10 (7pm)
$P'(t)$ (1000's ppl/hour)	7	11	8	4

Example 10: An amusement park opens its doors at 9AM ($t = 0$). The rate that people enter the park at time t is modeled by the function $P'(t)$, in 1000's of people per hour. Selected values of $P'(t)$ are given in the table above.



Use a trapezoidal approximation with the 3 subintervals indicated in the table to estimate the total number of people that entered the park from 9AM to 7PM. Write an integral expression for this value.

$$\int_0^{10} P'(t)\,dt \approx \frac{1}{2}\left[P'(0)+P'(2)\right]2 + \frac{1}{2}\left[P'(2)+P'(5)\right]3 + \frac{1}{2}\left[P'(5)+P'(10)\right]5$$

$$= \frac{1}{2}\left[(7+11)2 + (11+8)3 + (8+4)5\right]$$

$$= \frac{1}{2}\left[36+57+60\right]$$

$$= 76.5 \text{ thousands of people}$$

Over and Under Approximations

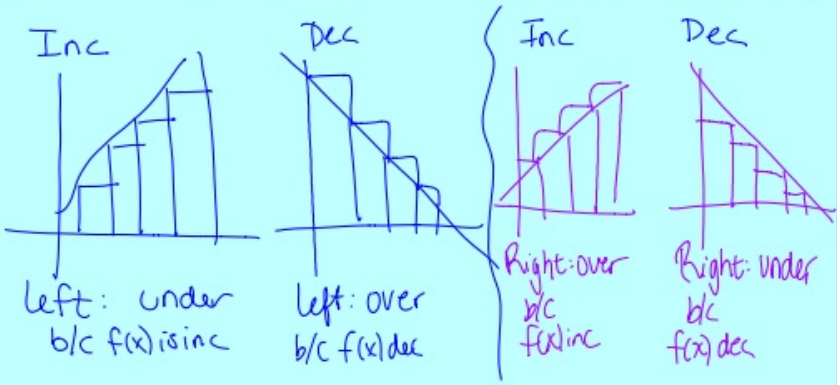
In calculus (and in the real world),we often need to make approximations or estimates. When we approximate a value, it is generally because we are not able to find the exact or true value. But even without knowing the exact value, we are often able to determine if our approximation was an overestimate or underestimate… including when we estimate a definite integral using Riemann sums or trapezoids.

Over vs. Under Approximations: Left and Right Riemann Sums

TYPE	IF…	Over or Under	Explanation
Left	$f(x)$ is increasing	Under	because $f(x)$ is increasing
	$f(x)$ is decreasing	Over	because $f(x)$ is decreasing
Right	$f(x)$ is increasing	Over	because $f(x)$ is increasing
	$f(x)$ is decreasing	Under	because $f(x)$ is decreasing

Over vs. Under Approximations: Midpoint Riemann Sums and Trapezoids

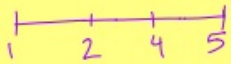
TYPE	IF…	Over or Under	Explanation
Midpoint	cons up $f''(x) > 0$	Under	because $f''(x) > 0$ on (a, b)
	cons down $f''(x) < 0$	Over	because $f''(x) < 0$ on (a, b)
Trapezoids	$f''(x) > 0$	Over	because $f''(x) > 0$ on (a, b)
	$f''(x) < 0$	Under	because $f''(x) < 0$ on (a, b)



Example 11: For $0 \le t \le 6$ seconds, a large glass is being filled with water at the rate $W'(t)$, measured in ounces per second. Selected values for decreasing function $W'(t)$ are given in the table above.

t (seconds)	1	2	4	5
$W'(t)$ (oz/sec)	2.3	1.7	1.2	0.3

a) Approximate $\int_1^5 W'(t)dt$ using a left Riemann sum with three subintervals as indicated in the table above.



$$\int_1^5 W'(t)dt \approx 1 \cdot W'(1) + 2 \cdot W'(2) + 1 \cdot W'(4)$$
$$= 2.3 + 2(1.7) + 1.2$$
$$= 6.9 \text{ oz}$$

b) Using correct units, interpret the meaning of $\int_1^5 W'(t)dt$ in context of the problem

The glass was filled with 6.9 oz of water from time $t = 1$ second to $t = 5$ seconds.

c) Is your answer from Part (a) an over or under approximation? Give a reason for your answer.

The answer in part (a) is an overestimate because $W'(t)$ is decreasing.